SRISHTI PALANI

srishtipalani.com | (413) 404-5828 | srishti@ucsd.edu

EDUCATION

University of California, San Diego September 2018 – Present
Ph.D. Cognitive Science

Mount Holyoke College, South Hadley, MA May 2018
B.A. *summa cum laude*, Computer Science, Psychology, Cognitive Neuroscience
Thesis: "*Who's There?": Designing Sensor-Aided Wearable Assistive Technology for the Blind*
Advisor: Barbara Lerner

AWARDS & HONORS

Grace Hopper Research Scholars' Mentor, Computing Research Association for Women 2019
Selected for Cornell, Maryland, Max Planck Pre-Doctoral Research School 2018
Outstanding Undergraduate Researcher (Finalist), Computing Research Association 2018
Best Undergraduate Research, Consortium for Computing Sciences in Colleges, North East 2018
Phi Beta Kappa Prize for Best Research, Phi Beta Kappa 2018
Fennema and Strahman Prize (Top Graduate), MHC Computer Science Department 2018
Academic Excellence Award (Top Junior, Senior), MHC Department of Psychology 2017, 2018
Diamond Award for Scholarship, National Residence Hall Honorary 2018
Grace Hopper Scholarship, Bank of America ($5,000 towards tuition) 2017
Mary Lyon Scholarship (merit scholarship) 2014-18

TECHNICAL SKILLS

Programming Languages: Python (pandas, numpy, scikit-learn, matplotlib, nltk, tensorflow); JavaScript (NodeJS, React, Angular), HTML/CSS, R, Swift
Platforms: Jupyter, Amazon Web Services, Microsoft Azure, iOS, Spark/Hadoop

RESEARCH EXPERIENCE

PhD Researcher, **Design Lab, UC San Diego** March 2019 – Present
Advisor: Steven Dow | Skills: Content Mining, NLP, Quantitative Analysis of User Interactions
Developed and evaluated novel computational techniques for implicitly mining relevance feedback and user context from user-generated content

Research Intern, **Microsoft Research** June – Sept 2019
Advisors: Merrie Morris, Adam Fourney | Skills: Eye-tracking, Search Log Analysis, User Studies
Recommended design requirements for making search tools more cognitively accessible, by conducting an eye-tracking study to investigate the differences in search behavior between searchers with and without dyslexia.

PhD Researcher (Rotations), **Design Lab, UCSD, San Diego** Sept 2018 – March 2019
Advisor: Jim Hollan, Scott Klemmer | Skills: User Studies, Contextual Inquiry, Qualitative Analysis

- Explored cognitive challenges and opportunities for personal information management
- Explored techniques for supporting creativity by harvesting existing expert content and recommending relevant examples to people in the context of their work.

Researcher, **Computer Science Department, Mount Holyoke College** Jan 2017 – May 2018
Advisor: Barbara Lerner | Skills: Context-Aware Embedded Systems, iOS Development, User Studies

- Designed a novel audio-haptic user interface that augments the visually impaired users' mind map of their surroundings by leveraging a smart environment
- Engineered a single-purpose wearable bracelet that lets the user identify persons in their surroundings
- Presented at CRA Outstanding Undergraduate Researcher Awards'18 (Finalist); and CCSCNE'18 (won Best Undergraduate Research Award)

PUBLICATIONS & PRESENTATIONS

Srishti Palani, Zijian Ding, Austin Nguyen, Andrew Chuang, Stephen MacNeil, Steven Dow. CoNotate: Suggesting Queries Based on Notes Promotes Knowledge Discovery. *In Proceedings of CHI'21: ACM CHI Conference on Human Factors in Computing Systems* (CHI '21). May 8-13, 2021, Virtual Conference. [26.3% Acceptance Rate]

Srishti Palani, Zijian Ding, Stephen MacNeil, Steven Dow. The "Active Search" Hypothesis: How Search Strategies Relate to Creative Learning. *In Proceedings of CHIIR'21: ACM SIGIR Conference on Human Information Interaction and Retrieval Online* (CHIIR'21). March 14-19, 2021, Virtual Conference.

Matin Yarmand, **Srishti Palani**, Scott Klemmer. Adjacent Display of Relevant Discussion Resolves Confusion when Learning Online. *Proceedings of the ACM on Human-Computer Interaction. (CSCW '21)*. November 2021, Virtual Conference.

Srishti Palani, Adam Fourney, Shane Williams, Kevin Larson, Irina Spiridonova, and Meredith Ringel Morris. 2020. An Eye Tracking Study of Web Search by People With and Without Dyslexia. *ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR '20)*, July 25-30, 2020, Virtual Event, China. [26% Acceptance Rate]

Srishti Palani, Barbara Lerner. "Who's There?": Designing Sensor-Aided Wearable Assistive Technology for the Blind. *Consortium for Computing Sciences in Colleges, North East (CCSCNE'18)*, March 2018, Manchester, New Hampshire. [Won Best Undergraduate Research Award]

Srishti Palani. Using Mathematical Modelling to Help the Agro-Based Economy in Drought-Prone Western India. *Women in Mathematics in New England (WIMIN'14)*, September 2014, Northampton, Massachusetts.

RESEARCH MENTORING

UCSD Masters' Student: *Zijian Ding*
UCSD Undergraduate Students: *Brendan Jew, Austin Nguyen*
CMU Human Computer Interaction Institute Undergraduate Student: *Andrew Chuang*

INDUSTRY EXPERIENCE

Intern, **Operational Intelligence Group, Bank of America** June - August 2017

- Developed automated machine learning algorithms that track dynamic dependencies between backend financial transaction applications; and learn and predict patterns of failures in the technological architecture supporting mobile/online banking and trading systems.

- Learned and applied time series modeling and classification machine learning algorithms and implemented Dynamic Time Warping to extract causal predictions
- Coordinated with cross-cultural, cross-disciplinary teams and recommended the design of smart environments as a move towards card-less banking and bridging the gap between online and traditional banking

Data Science Intern, **Cognitive Computing & Analytics Group, IBM** May-August 2016

- Contributed to developing and testing Alchemy Language: Watson 2.0's Natural Language Processing API as part of a large interdisciplinary international team
- Constructed a program to use the API to extract sentiment, keywords, and high-level concepts from social media & search data in an area to then predict disease outbreak
- Established cloud computing architecture to analyze clickstream data in real-time & perform market segmentation

TEACHING EXPERIENCE

Lead Teaching Assistant, **UCSD Cognitive Science Department** June 2020 – Present

- Assist professors, instructors, and TAs across the department with online lesson plans and technology for remote learning; and organize discussion roundtables to share experiences
- Liaison between the department and the School of Social Sciences

Teaching Assistant, **Startup Studio, UCSD Cognitive Science** Fall 2020
Teaching Assistant, **Civic Design**, **UCSD Cognitive Science** Winter 2020
Teaching Assistant, **Social Computing**, **UCSD Cognitive Science** Fall 2019
Teaching Assistant, **Prototyping**, **UCSD Cognitive Science** Spring 2019

Teaching Assistant, **Human Computer Interaction**, **UCSD Cognitive Science** Winter 2019

- In the above positions, I assisted Professors Steven Dow and Jim Hollan in designing project-based class for 50-150 students, moderated discussions, lead a few lectures, helped with grading and mentor students

Teaching Assistant, **Intelligent Information Retrieval, MHC Computer Science** Spring 2018

- Assisted Professor Lisa Ballesteros in designing project-based class for ~20 students by the end of which students had to develop their own custom information retrieval systems.
- Moderated discussions and helped students debug their individual projects' codebases in office hours

Teaching Assistant, **Programming Language Design, MHC Computer Science** Fall 2017

- Assisted Professor Valerie Barr in designing project-based class for ~30 students by the end of which students had to develop their own custom programming language. Moderated discussions and helped students debug their individual projects' codebases in office hours

LEADERSHIP EXPERIENCE

Head of Outreach, **UCSD Graduate Women in Computing** May 2019 - Present

- Organized two conferences for local schools, and three speaker series for undergraduates and graduates to facilitate interactions with between students and mentors from under-represented backgrounds in computing

Graduate Mentorship Coordinator, **UCSD Cognitive Science** August 2019 - June 2020

- Creating a trans-disciplinary peer mentorship program between graduate students in computing fields across UCSD

Senior Community Advisor, **MHC Office of Residential Life** August 2015 – May 2018

- Fostered an inclusive and friendly community in the largest community on campus of 200+ students; Guided 8 Community Advisors to be role models, peer counselors, event planners, crisis responders
- Directed inclusive programming for the Mary Woolley Floor – our LGBTQIA+ focused living learning community

Harriet Newhall Fellow, **MHC Office of Admissions** August 2017 – May 2018

- Interacted with and interviewed prospective students; aided in the planning and execution of admission special events

Member, Student Advisory Committee, **MHC Department of Computer Science** Fall 2016

- Engaged with faculty research, attended talks, and provided feedback during recruitment of senior faculty member to MHC CS Department

Co-Founder & Chair, **MHC Girls in Tech Conference** March 2016 – May 2017

- Spearheaded a conference geared for high school girls from suburban New England to teach them about the versatility of the tech industry through lightning talks, workshops, and mentorship; Led a team of 14 board members, 50+ student volunteers and mentors
- Coordinated over $8,000 in funding and sponsorship
- As Head of Outreach and Sponsorship in 2015-2016: helped found the conference, successfully procured over 15 sponsors and 2 award grants of over $15,000; invited and organized 10+ speakers and workshop leaders over the next 2 years

Board Member, **HackHolyoke'14 – '16** September 2014 - March 2016

- Helped organize the first women's college hackathon, the first hackathon to achieve 50:50 gender ratio; Organized 60+ volunteers and arranged for resources for 400+ hackers over 2 hackathons

Member, **Varsity Squash Team, MHC** September 2014 - September 2016